

Checkpoint 0:

```
appdb=> \conninfo
Connection Information
Parameter | Value
-----+-----
Database | appdb
Client User | app
Host | localhost
Host Address | ::1
Server Port | 5432
Options |
Protocol Version | 3.0
Password Used | true
GSSAPI Authenticated | false
Backend PID | 14408
SSL Connection | false
Superuser | off
Hot Standby | off
(13 rows)
```

1.1 Create Tables:

```
appdb=> \dt
List of tables
Schema | Name | Type | Owner
-----+-----+-----+-----
public | hero | table | app
public | team | table | app
(2 rows)
```

```
appdb=> \d hero
Table "public.hero"
Column | Type | Collation | Nullable |
Default |
-----+-----+-----+-----+-----
id | integer | | not null | nextval('
hero_id_seq'::regclass)
name | character varying | | not null |
age | integer | | |
team_id | integer | | |
Indexes:
    "hero_pkey" PRIMARY KEY, btree (id)
Foreign-key constraints:
    "hero_team_id_fkey" FOREIGN KEY (team_id) REFERENCES team(i
d)
```

1.2 Insert and query

```
appdb=> INSERT INTO team (name, headquarters)
appdb-> VALUES ('Avengers', 'New York'), ('X-Men', 'Westchester
');
INSERT 0 2
appdb=>
appdb=> INSERT INTO hero (name, age, team_id)
appdb-> VALUES ('Tony', 45, 1), ('Natasha', 35, 1), ('Logan', 1
50, 2), ('Peter', 16, NULL);
INSERT 0 4
appdb=>
appdb=> SELECT * FROM hero;
 id |  name  |  age  | team_id
----+-----+-----+-----
  1 | Tony   |  45   |      1
  2 | Natasha |  35   |      1
  3 | Logan  | 150   |      2
  4 | Peter  |  16   |
(4 rows)
```

Task 1. Write SQL queries for:

1. All heroes aged 18 or older, oldest first.

```
appdb=> SELECT * FROM hero WHERE age >= 18 ORDER BY age DESC;
 id |  name  |  age  | team_id
----+-----+-----+-----
  3 | Logan  | 150   |      2
  1 | Tony   |  45   |      1
  2 | Natasha |  35   |      1
(3 rows)
```

2. Each hero's name next to their team's name (heroes without a team must also appear).

```
appdb=> SELECT hero.name AS hero_name, team.name AS team_name
appdb-> FROM hero LEFT JOIN team ON hero.team_id = team.id;
 hero_name | team_name
-----+-----
 Tony      | Avengers
 Natasha   | Avengers
 Logan     | X-Men
 Peter     |
(4 rows)
```

3. The number of heroes per team.

```
appdb=> SELECT team.name, COUNT(hero.id) AS hero_count
appdb-> FROM team LEFT JOIN hero ON team.id = hero.team_id
appdb-> GROUP BY team.name;
  name      | hero_count
-----+-----
X-Men       |          1
Avengers    |          2
(2 rows)
```

4. The 2nd page of heroes when a page has 2 rows, ordered by id .

```
appdb=> SELECT * FROM hero ORDER BY id LIMIT 2 OFFSET 2;
 id | name  | age | team_id
----+-----+----+-----
  3 | Logan | 150 |      2
  4 | Peter |  16 |
(2 rows)
```

1.3 Let the database protect your data

The error message:

```
appdb=> INSERT INTO team (name, headquarters) VALUES ('Avengers', 'Los Angeles');
ERROR:  duplicate key value violates unique constraint "team_name_key"
DETAIL:  Key (name)=(Avengers) already exists.
appdb=> INSERT INTO hero (name, team_id) VALUES ('Ghost', 99);
ERROR:  insert or update on table "hero" violates foreign key constraint "hero_team_id_fkey"
DETAIL:  Key (team_id)=(99) is not present in table "team".
appdb=> INSERT INTO hero (age) VALUES (30);
ERROR:  null value in column "name" of relation "hero" violates not-null constraint
DETAIL:  Failing row contains (6, null, 30, null).
appdb=> DELETE FROM team WHERE id = 1;
ERROR:  update or delete on table "team" violates foreign key constraint "hero_team_id_fkey" on table "hero"
DETAIL:  Key (id)=(1) is still referenced from table "hero".
```

Checkpoint 1:

```
appdb=> DROP TABLE hero;
DROP TABLE
appdb=> DROP TABLE team;
DROP TABLE
appdb=> \dt
Did not find any tables.
appdb=> \q
```

Checkpoint 2:

```
(.venv) (base) PS C:\Users\DELL\hero-api> $env:DATABASE_URL = "
postgresql+psycpg://app:secret@localhost:5432/appdb"
(.venv) (base) PS C:\Users\DELL\hero-api> echo $env:DATABASE_UR
L
postgresql+psycpg://app:secret@localhost:5432/appdb
```

3.1

```
from sqlmodel import Field, Relationship, SQLModel

class Team(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True, unique=True)
    headquarters: str

    heroes: list["Hero"] = Relationship(back_populates="team")

class Hero(SQLModel, table=True):
    id: int | None = Field(default=None, primary_key=True)
    name: str = Field(index=True)
    age: int | None = None
    secret_name: str
    team_id: int | None = Field(default=None, foreign_key="team.i

    team: Team | None = Relationship(back_populates="heroes")
```

Checkpoint 3:

```
(.venv) (base) PS C:\Users\DELL\hero-api> python -c "from app.models import Hero, HeroPublic; print(Hero.__tablename__)"
hero
```

Checkpoint 4:

```
(.venv) (base) PS C:\Users\DELL\hero-api> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" "postgresql://app:secret@localhost:5432/appdb" -c "\dt"
>>
          List of tables
 Schema | Name | Type  | Owner
-----+-----+-----+-----
 public | hero | table | app
 public | team | table | app
(2 rows)
```

```
(.venv) (base) PS C:\Users\DELL\hero-api> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" "postgresql://app:secret@localhost:5432/appdb" -c "\d hero"
Table "public.hero"
 Column |          Type          | Collation | Nullable |          Default
-----+-----+-----+-----+-----
 name   | character varying     |           | not null |
 age    | integer               |           |          |
 team_id | integer               |           |          |
 id      | integer               |           | not null | nextval('hero_id_seq'::regclass)
 secret_name | character varying |           | not null |
Indexes:
    "hero_pkey" PRIMARY KEY, btree (id)
```

5.1 Test

curl -X POST localhost:8000/heroes -H "Content-Type: application/json" -d '{"name": "Peter", "age": 16, "secret_name": "Spider-Man"}'

```
INFO: 127.0.0.1:65025 - "POST /heroes HTTP/1.1" 201 Created
2026-10-02 23:53:17,442 INFO sqlalchemy.engine.Engine ROLLBACK
2026-10-02 23:55:22,164 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-02 23:55:22,164 INFO sqlalchemy.engine.Engine SELECT hero.name, hero.age, hero.team_id, hero.id, hero.secret_name
FROM hero ORDER BY hero.id
LIMIT %(param_1)s::INTEGER OFFSET %(param_2)s::INTEGER
2026-10-02 23:55:22,164 INFO sqlalchemy.engine.Engine [generated in 0.00041s] {'param_1': 10, 'param_2': 0}
```

curl "localhost:8000/heroes?limit=10"

```
INFO:      127.0.0.1:55244 - "GET /heroes?offset=0&limit=10 HTTP/1.1" 200 OK
2026-10-02 23:55:22,176 INFO sqlalchemy.engine.Engine ROLLBACK
2026-10-02 23:58:25,784 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-02 23:58:25,784 INFO sqlalchemy.engine.Engine SELECT hero.name AS hero_name, hero.age AS hero_age, hero.team_id AS hero_team_id, hero.id AS hero_id, hero.secret_name AS hero_secret_name
FROM hero
WHERE hero.id = %(pk_1)s::INTEGER
2026-10-02 23:58:25,784 INFO sqlalchemy.engine.Engine [generated in 0.00039s] {'pk_1': 1}
INFO:      127.0.0.1:40303 - "GET /heroes/1 HTTP/1.1" 200 OK
```

curl -X PATCH localhost:8000/heroes/1 -H "Content-Type: application/json" -d '{"age": 17}'

```
INFO:      127.0.0.1:59839 - "PATCH /heroes/1 HTTP/1.1" 200 OK
2026-10-03 00:09:27,415 INFO sqlalchemy.engine.Engine ROLLBACK
2026-10-03 00:09:34,331 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:09:34,331 INFO sqlalchemy.engine.Engine SELECT hero.name AS hero_name, hero.age AS hero_age, hero.team_id AS hero_team_id, hero.id AS hero_id, hero.secret_name AS hero_secret_name
FROM hero
WHERE hero.id = %(pk_1)s::INTEGER
2026-10-03 00:09:34,333 INFO sqlalchemy.engine.Engine [cached since 668.5s ago] {'pk_1': 1}
2026-10-03 00:09:34,337 INFO sqlalchemy.engine.Engine DELETE FROM hero WHERE hero.id = %(id)s::INTEGER
2026-10-03 00:09:34,337 INFO sqlalchemy.engine.Engine [generated in 0.00037s] {'id': 1}
2026-10-03 00:09:34,343 INFO sqlalchemy.engine.Engine COMMIT
```

curl -i localhost:8000/heroes/999

```
INFO:      127.0.0.1:50597 - "GET /heroes/999 HTTP/1.1" 404 Not Found
[]
```

curl -i -X DELETE localhost:8000/heroes/1

```
INFO:      127.0.0.1:61197 - "DELETE /heroes/1 HTTP/1.1" 204 No Content
2026-10-03 00:11:06,873 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:11:06,873 INFO sqlalchemy.engine.Engine SELECT hero.name AS hero_name, hero.age AS hero_age, hero.team_id AS hero_team_id, hero.id AS hero_id, hero.secret_name AS hero_secret_name
FROM hero
WHERE hero.id = %(pk_1)s::INTEGER
2026-10-03 00:11:06,873 INFO sqlalchemy.engine.Engine [cached since 761.1s ago] {'pk_1': 999}
2026-10-03 00:11:06,873 INFO sqlalchemy.engine.Engine ROLLBACK
```

Checkpoint 5:

- Create 3 heroes and stop session

```
INFO:      127.0.0.1:65192 - "POST /heroes HTTP/1.1" 201 Created
2026-10-03 00:18:41,935 INFO sqlalchemy.engine.Engine ROLLBACK
2026-10-03 00:18:51,118 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:18:51,119 INFO sqlalchemy.engine.Engine INSERT INTO hero (name, age, team_id, secret_name) VALUES (%(name)s::VARCHAR, %(age)s::INTEGER, %(team_id)s::INTEGER, %(secret_name)s::VARCHAR) RETURNING hero.id
2026-10-03 00:18:51,119 INFO sqlalchemy.engine.Engine [cached since 1534s ago] {'name': 'Tony', 'age': 35, 'team_id': None, 'secret_name': 'Iron Man'}
2026-10-03 00:18:51,119 INFO sqlalchemy.engine.Engine COMMIT
2026-10-03 00:18:51,121 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:18:51,121 INFO sqlalchemy.engine.Engine SELECT hero.name, hero.age, hero.team_id, hero.id, hero.secret_name
```

```

INFO:      127.0.0.1:64954 - "POST /heroes HTTP/1.1" 201 Created
2026-10-03 00:18:51,130 INFO sqlalchemy.engine.Engine ROLLBACK
2026-10-03 00:18:59,774 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:18:59,774 INFO sqlalchemy.engine.Engine INSERT INTO hero (name, age, team_id, secret_name) VALUES (%(name)s::VARCHAR,
%(age)s::INTEGER, %(team_id)s::INTEGER, %(secret_name)s::VARCHAR) RETURNING hero.id
2026-10-03 00:18:59,775 INFO sqlalchemy.engine.Engine [cached since 1542s ago] {'name': 'Steve', 'age': 30, 'team_id': None, 'secret
_name': 'Captain America'}
2026-10-03 00:18:59,776 INFO sqlalchemy.engine.Engine COMMIT
2026-10-03 00:18:59,781 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:18:59,782 INFO sqlalchemy.engine.Engine SELECT hero.name, hero.age, hero.team_id, hero.id, hero.secret_name

```

```

INFO:      127.0.0.1:59205 - "POST /heroes HTTP/1.1" 201 Created
2026-10-03 00:18:59,785 INFO sqlalchemy.engine.Engine ROLLBACK
INFO:      Shutting down
INFO:      Waiting for application shutdown.
INFO:      Application shutdown complete.
INFO:      Finished server process [36148]
INFO:      Stopping reloader process [28324]
(.venv) (base) PS C:\Users\DELL\hero-api>

```

- Start session again and call GET /heroes

```

INFO:      Application startup complete.
INFO:      127.0.0.1:51540 - "GET /docs HTTP/1.1" 200 OK
INFO:      127.0.0.1:51540 - "GET /openapi.json HTTP/1.1" 200 OK
2026-10-03 00:21:57,968 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-10-03 00:21:57,977 INFO sqlalchemy.engine.Engine SELECT hero.name, hero.age, hero.team_id, hero.id, hero.secret_name
FROM hero ORDER BY hero.id
LIMIT %(param_1)s::INTEGER OFFSET %(param_2)s::INTEGER
2026-10-03 00:21:57,979 INFO sqlalchemy.engine.Engine [generated in 0.00068s] {'param_1': 10, 'param_2': 0}
INFO:      127.0.0.1:49711 - "GET /heroes?offset=0&limit=10 HTTP/1.1" 200 OK
2026-10-03 00:21:57,986 INFO sqlalchemy.engine.Engine ROLLBACK

```

200

Response body

```

[
  {
    "name": "Peter",
    "age": 19,
    "team_id": null,
    "id": 2
  },
  {
    "name": "Tony",
    "age": 35,
    "team_id": null,
    "id": 3
  },
  {
    "name": "Steve",
    "age": 30,
    "team_id": null,
    "id": 4
  }
]

```

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Checkpoint 6:

Duplicate team → 409

201

Response body

```
{  
  "name": "Avengers",  
  "headquarters": "New York",  
  "id": 1  
}
```



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Response headers

```
content-length: 52  
content-type: application/json  
date: Sat,03 Oct 2026 07:14:07 GMT  
server: uvicorn
```

409

Undocumented

Error: Conflict

Response body

```
{  
  "detail": "Team already exists"  
}
```



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Response headers

```
content-length: 32  
content-type: application/json  
date: Sat,03 Oct 2026 07:15:25 GMT  
server: uvicorn
```

Responses

Code

Description

Links

Unknown team id → 404

404
Undocumented

Error: Not Found

Response body

```
{  
  "detail": "Team not found"  
}
```

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Response headers

```
content-length: 27  
content-type: application/json  
date: Sat, 03 Oct 2026 07:42:33 GMT  
server: uvicorn
```

```
INFO:      127.0.0.1:64092 - "PATCH /heroes/2 HTTP/1.1" 404 Not Found  
2026-10-03 14:44:55,725 INFO sqlalchemy.engine.Engine BEGIN (implicit)  
2026-10-03 14:44:55,725 INFO sqlalchemy.engine.Engine SELECT hero.name AS hero_name, hero.age AS hero_age, hero.team_id AS hero_team  
_id, hero.id AS hero_id, hero.secret_name AS hero_secret_name  
FROM hero  
WHERE hero.id = %(pk_1)s::INTEGER  
2026-10-03 14:44:55,725 INFO sqlalchemy.engine.Engine [cached since 58.79s ago] {'pk_1': 2}
```

The filters combine correctly.

Request URL

http://127.0.0.1:8000/heroes?offset=0&limit=10&min_age=18&team_id=1

Server response

Code

Details

200

Response body

```
[  
  {  
    "name": "Thor",  
    "age": 30,  
    "team_id": 1,  
    "id": 5  
  }  
]
```

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Checkpoint 7:

```
(.venv) (base) PS C:\Users\DELL\hero-api> python -c "import os; print(repr(os.getenv('DATABASE_URL')))"  
None  
& "C:\Program Files\PostgreSQL\18\bin\psql.exe" "postgresql://app:secret@localhost:5432/ap  
pdb" -c "\dt") PS C:\Users\DELL\hero-api>  
List of tables  
Schema | Name | Type | Owner  
-----+-----+-----+-----  
public | hero | table | app  
public | heromissionlink | table | app  
public | mission | table | app  
public | team | table | app  
(4 rows)
```

```
(.venv) (base) PS C:\Users\DELL\hero-api> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" "postgresql://app:secret@localhost:5432/ap
pdb" -c "\d heromissionlink"
      Column | Type   | Collation | Nullable | Default
-----+-----+-----+-----+-----
 hero_id    | integer |           | not null |
 mission_id | integer |           | not null |
Indexes:
    "heromissionlink_pkey" PRIMARY KEY, btree (hero_id, mission_id)
Foreign-key constraints:
    "heromissionlink_hero_id_fkey" FOREIGN KEY (hero_id) REFERENCES hero(id)
    "heromissionlink_mission_id_fkey" FOREIGN KEY (mission_id) REFERENCES mission(id)
```

After assigning hero 1 to mission 1 twice, heromissionlink contains exactly one row (1, 1)

```
(.venv) (base) PS C:\Users\DELL\hero-api> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" "postgresql://app:secret@localhost:ap
pdb" -c "SELECT * FROM heromissionlink;"
 hero_id | mission_id
-----+-----
      1 |          1
(1 row)
```

Checkpoint 8:

200

Response body

```
[
  {
    "name": "Iron Man",
    "age": 35,
    "team_id": 1,
    "id": 1
  },
  {
    "name": "Captain America",
    "age": 100,
    "team_id": 1,
    "id": 3
  },
  {
    "name": "Thor",
    "age": 1500,
    "team_id": 1,
    "id": 4
  },
  {
    "name": "Hulk",
    "age": 40,
    "team_id": 1,
    "id": 5
  },
  {
    "name": "Black Widow",
    "age": 35,
```



Download

Request URL

http://127.0.0.1:8000/heroes/1/missions

Server response

Code	Details
200	Response body <pre>[{ "id": 1, "title": "Battle of Sokovia" }, { "id": 2, "title": "Battle of New York" }]</pre> Response headers

Checkpoint 9:

```
(.venv) (base) PS C:\Users\DELL\hero-api> alembic history
<base> -> 9a4871c40aff (head), add hero.power
(.venv) (base) PS C:\Users\DELL\hero-api> alembic current
INFO [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO [alembic.runtime.migration] Will assume transactional DDL.
9a4871c40aff (head)
```